EXCEL ENGINEERING COLLEGE (AUTONOMOUS) - DEPARTMENT OF ECE

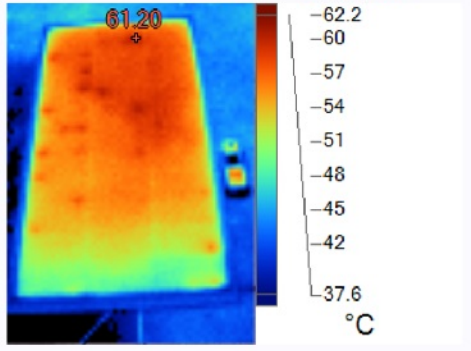
SOLAR PANEL DEFECT DETECTION & DIAGNOSTIC REPORT (MATLAB FUSION PIPELINE)

Panel ID: Panel1 Dirt | Inspection Date: 15-Oct-2026 10:10:29 | Analysis Method: Dual Thermal-Visual Feature Fusion

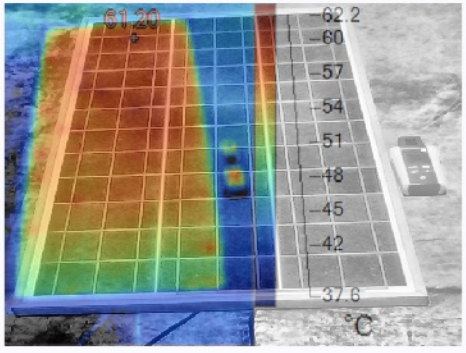
(A) Visual (RGB) Inspection



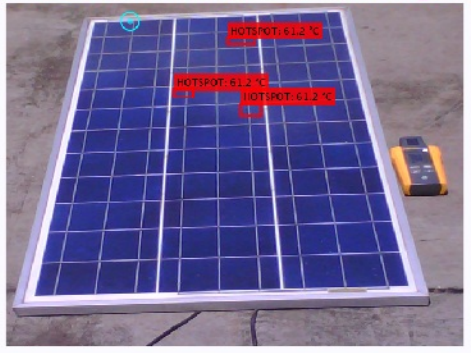
(B) Thermal Thermography (T_{max}: 61.2 °C)



(C) imfuse Thermal-Visual Composite



(D) Defect Localization & Bounding Boxes



DIAGNOSTIC SUMMARY & THERMAL METRICS

- Primary Defect Identified : Hotspot / Cell Overheating
- Peak Temperature (T_{max}) : 61.15 °C (Ambient Mean: 51.95 °C)
- Temperature Rise (Δ T) : +9.20 °C
- Affected Panel Surface : 14.5 %
- Severity Rating : [CRITICAL]

STATUS CRITICAL

ENGINEERING MAINTENANCE RECOMMENDATION:

Critical localized thermal hotspot. High risk of EVA browning / backsheet burn-through. Isolate string and replace cell/module immediately.